CAPSTONE PROJECT DETAILED PLAN--(last updated Spring 2026)

PERMANENT INSTRUCTIONS: Because instructors, advisors, mentors, and peers will be looking at this file throughout the semester to provide feedback, please do not work in any other documents than this one. As deadlines occur for INDS 480, students will be assessed (graded) on what is in this document alone. Before enrollment into INDS490, a complete version of this document must be submitted and approved by the INDS 480 instructor and a project protocol submitted to the IRB.

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List of degree mentor names, one per line, each name followed by their official title and email address[1]

Mentor Name	**Title**	**Email Address**
Stephen Freeland	Director, Individualized Study	freeland@umbc.edu
Dillon Mahmoudi	Graduate Program Director, GES and GIS	dillonm@umbc.edu

[1] For UMBC faculty and staff, be sure to use the UMBC directory to find this title with certainty. For all others, please make it an explicit point to ask/verify at your next meeting with the mentor.

CAPSTONE PROJECT DETAILED PLAN
Eight sections, five appendices

1. Project Title: Provide a working title for the project in 15 words or less
Identifying where biosignatures are most likely to be preserved in the greater Jezero Crater

2. Project description: In approximately 300-400 words, explain to your reader what your topic is, a brief background for understanding the topic, your research question, your thesis statement, and the methods (literature analysis, interviews, surveys, creative composition, etc.) you will use to address your research question.

A. What topic or issue does this capstone project address?
B. Provide some background (history; definitions; key concepts; theories; people working on this topic/problem).
C. Research Question
D. Thesis statement[2] - tentative answer to your research question
E. Brief overview of study components and methods

My project is going to answer the question: "How can geochemical, spatial, and environmental data be used to determine the likelihood of biosignature preservation potential within Jezero Crater?" I will be exploring different datasets and imagery to create maps to visualize and identify where potential future missions could find the most likely samples that would contain preserved amino acids and protein structures.

I will be focusing on Jezero Crater, an ancient Martian lake basin and the current exploration site of NASA's *Perseverance* rover. Jezero is considered to be a high-priority astrobiological target because it once contained a standing body of water, connecting rivers, and a sedimentary delta system. On Earth, similar environments are known to concentrate and preserve organic materials under the right conditions. In astrobiology, a biosignature refers to any measurable indicator of present or past life. This can include anything from the distribution of elements, to monomers such as amino acids, cell-like structures, or changes in behavior due to selective pressures (Ladder of Life Detection, NASA). The preservation of such compounds is heavily determined by the environmental context, including mineralogy, redox conditions, radiation/erosion exposure, and burial history.

My thesis statement is that regions within the crater that combine fine-grain sediment deposition, hydrated mineral phases, and low deposition alteration will show the highest potential for preservation of biosignatures, specifically amino acids and protein related structures.

To address my research question, I will use a literature analysis alongside spatial and geochemical data interpretation. I will use data from open-source databases, such as NASA and USGS to identify relevant environmental indicators. I will then synthesize data models in ArcGIS Pro to show the variation in potential across the crater. The final outcome will be a set of predictive maps identifying high-probability sampling zones, which can inform future mission planning and contribute to broader strategies in biosignature detection on Mars.

3. Study design: How will your study investigate your research question(s)?: In 300-400 words provide a detailed description of the methods/techniques you will use in your capstone project and how you will use them. Some examples of methods include literature analysis, surveys, interviews, prototype design, reflection paper, experiments, case studies, creative composition, and discourse analysis.

1. Literature analysis
    a. I will continue my literature review from this semester and add sources relating to **methods** of mapping

[2] Thesis here is meant in the sense of Merriam Webster " **2b:** a position or proposition that a person (such as a candidate for scholastic honors) advances and offers to maintain by argument." Your project demonstrates your thesis, whether through words alone, or through a mixture of words and other types of deliverables. For further reading, see for example https://owl.purdue.edu/owl/general_writing/the_writing_process/thesis_statement_tips.html

Mars, as well as **databases** that I plan to use and why I will be using them. I want to inform myself of the processes other researchers have done so I am able to come up with a plan for my personal mapping and analysis process. I will also research similar maps (e.g. a geomorphological map) and include them in the review.

2. Data analysis
    a. In ArcGIS, I will be analyzing shapefiles and imagery to determine sites of highest biosignature preservation potential based on a few factors, namely geomorphology and the distribution of organic compounds within the crater. I will first be taking a look at what the Mars 2020 *Perseverance* rover has collected, and then extrapolating this data to hypothesize where similar samples would be collected elsewhere in the area of interest.
3. Map creation
    a. I will be making many different types of maps of Jezero Crater. I will focus on the *Perseverance* path and create a few maps of the area it has traveled thus far, and then zoom out to create the same maps for the entire area. My thoughts right now are making a geomorphology map, a chemical distribution map, and finally, an interpolation map that combines these two factors into where I hypothesize that amino acid or peptide structures are most likely to be preserved.
4. Research paper
    a. As I am doing data analysis and map creation, I will simultaneously be writing a research paper that outlines my thought processes, methods, and rationale behind my analyses.
5. Digital portfolio
    a. Finally, I will be making a digital portfolio piece that I can show to future employers or during my grad school application process. I am not sure yet what shape that will take. ArcGIS has an interactive feature called StoryMaps that I could use, but I am not sure if that will be able to fully cover the scope of my project. I am leaning towards creating a website that I can then go on and use for work that I've done and will do relating to my interest in astrobiology, chemistry, and GIS.

4. Annotated Bibliography: Your annotated bibliography will form the basis for your literature review.

A. Please choose and apply consistently a citation format that is appropriate to the academic culture of your project (for example, a project which is heavily reliant on Psychology might well choose "APA format", whereas one focused on molecular biology could choose from a wide variety of formats associated with different journals or funding agencies.) In all cases, please consult with mentors about this choice, and make your choice explicit below.
B. Chosen citation style:
C. Please **organize your references by discipline or applied field**.
D. Include a link to an alphabetized bibliography.
E. Compare this list to the areas in your concept map. Is each area of your map "well informed"?

ANNOTATED BIBLIOGRAPHY

5. Literature Review including current State of the Debate on the topic: It is useful to identify foundational works[3] in your project and to place them in conversation with one another. This conversation is your State of the Debate.

LITERATURE REVIEW DRAFT 1
LITERATURE REVIEW DRAFT 2

6. How will this project be delivered? *Thesis capstones are usually delivered in the form of a single, long paper (e.g.*

[3] Foundational works are not necessarily written; works could be in the performing arts or an early version of a new technology.

MSWord or PDF) Portfolios capstones are usually delivered at the completion of the project in the form of a web-site comprising appropriate documentation/links to the different components along with an introduction/overview to the project as a whole. The elements of a capstone project may be delivered in many different ways.
My project will be delivered in the form of a research paper. Within my paper, I plan to outline the process I went through in identifying what kinds of sites I would be looking for, the methods in which I analyze the data, and finally, the process that I will use to create maps. Within the paper, I will create several different maps of the Martian surface in ArcGIS Pro to visually show what I plan to do. I also would like to create a final research poster to accompany my paper as something that I could present easier and be able to put in my portfolio to show to employers in the future.

7. Detailed Timeline: *Begin your timeline planning by making a list of all the components of your project: a review of the literature, any creative components, interviews and when you plan to conduct them, your presentation, etc. Then, estimate roughly how long you believe each task will take. Next, using the list of required INDS deliverables (see below) and dates as an outline, draft an estimated sequence of your project tasks. Some helpful tips: a) Begin your timeline with tomorrow's date (including such tasks as obtaining IRB approval, etc.); b) build in extra time for revisions and/or "iterations" where you revisit an earlier section of your report; c) think of your 490 capstone semester as a series of two week "sprints" - when you consider this, you will see that you have about six sessions of content creation.*

Detailed Capstone Timeline

8. IRB Approval or Exemption: *Each student is required to submit their project to IRB for evaluation. In this section provide proof of that submission such as quoted email correspondence and/or PDF letters from the IRB and direct evidence of any formal permissions granted. Note that no student may be enrolled into INDS 490: Capstone Project unless proof of protocol submission to the IRB is provided.*
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Institutional Review Board
FWA #00000069

University of Maryland, Baltimore County
1000 Hilltop Circle
Baltimore, MD 21250

PHONE: 410-455-2737
EMAIL: irb@umbc.edu

Date: May 4, 2026
To: Sofia Helene Dominesey, Dr. Stephen Freeland
Department: Individualized Studies

Notice of Action: Not Human Subjects Research Determination

Kuali Protocol #: 2210
Kuali Protocol Title: Identifying sites of highest biosignature preservation potential on Mars

This letter is to acknowledge that the Office of Research Protections and Compliance and the UMBC IRB have reviewed the information provided in the above-mentioned protocol **Kuali 2210** and has determined that the submission **does not** require further review. This determination has been made with the understanding that the proposed project does not involve the following: a systematic investigation designed to develop or contribute to generalizable knowledge 46.102(7)(l) **OR** a human subject 46.102(e)(1).

This determination applies only to the activities described in the protocol listed.

If there are plans to make changes or if there are further questions as to whether these activities are human subject research, please do not hesitate to contact the ORPC at 5-2737 or irb@umbc.edu.

Capstone Project <u>Required</u> Elements: In order to successfully complete the capstone project, each student must include the following elements as part of their final deliverables for INDS 490. Some of these elements will be started during INDS 480.

1. Review of relevant published literature
2. Bibliography
3. Capstone presentation with visuals

Appendix 1: Project alignment with degree plan courses and co-curricular activities

In this section of 400-500 words, describe between three (3) and five (5) courses and any co-curricular activities that are of particular importance to your degree plan. Be sure to include a discussion of the important works[4], theories, methods, and/or techniques you encountered in each of the classes and activities you identify and the role you see them playing in your capstone project. For the important works, identify the disciplines or applied fields in which they were published and state clearly what each one contributes to your understanding.

GES 286 and GES 386 (Geographic Information Systems) taught me the technical foundations that I need for working with spatial data. In these two courses I learned how to manage many different types of datasets, as well as manipulate them to get certain types of information. In addition to this, I also learned how to present data and information aesthetically and effectively on maps and posters. These courses prepared me for my capstone project by equipping me with these skills, as well as with the soft skills of learning how to use different types of software without much background or support, which will be especially useful, as I play to map data from Mars on different software.

In **GES 381 (Remote Sensing)**, I learned how to interpret and analyze data from satellite imagery and other forms of remote sensing techniques. I learned how Earth-observing satellites work and collect data via classification techniques, electromagnetic radiation applications, and spectral signatures. I also learned technical skills such as atmospheric correction and extracting surface data from data taken from a distance. These types of techniques are essential for interpreting Mars data, as there are many Mars-observing satellites with a treasure tribe of information to analyze. The ability to interpret satellite imagery and datasets will directly inform how I will identify sites on Mars containing possible biosignatures and conditions for life.

CHEM 490 (Advanced Spectroscopy) deepened my understanding of how electromagnetic radiation interacts with molecules themselves and why we observe certain patterns due to molecular properties. I studied molecular orbital theory, electronic transitions, vibrational modes, and selection rules in regards to various types of spectroscopies. This class helped me understand why molecules and organic compounds behave the way that they do in terms of the spectral signatures they emit, transmit, and absorb. In my capstone, I will apply what I learned to not only interpret visually, but mechanistically as well, understanding what types of bonds and molecules are present by reading spectroscopic data.

INDS 430 (Astrobiology) served as the backbone of my entire degree plan. I learned the basics of everything relating to my field in this course, from the instruments that detect biosignatures to planetary geology and geography to history of the universe and so much more. Having a good foundation in the field of astrobiology as a whole will strengthen the ability of me to excel in my capstone by having the skills to interpret my findings in a broad sense. The interdisciplinary perspective that this class gave me shaped my capstone's central research question that involves bringing two different subfields together.

GES 302 (Paleoclimates) introduced me to different ways of theorizing what Earth's surface and climate was before technology was there to observe it actively. Although it was focused on Earth, I learned the skills needed to apply the same techniques to Mars in order to theorize why and how Mars lost its atmosphere, as well as how the hydrology of the red planet evolved over time. Understanding climate transitions on our own planet enables me to contextualize patterns and data that I will encounter from Mars.

[4] "Important works" include foundational publications, works or new technology that you found especially helpful, works of art, technical manuals, methodologies, etc.

Appendix 2: Project alignment with career goals

Post graduate goals: How will this project help you to understand the relevant issues in your professional field? What kinds of problems are being addressed by your professional field? What are the issues and what are your role models working on?

My capstone project directly aligns with my post-graduation goals and the type of work I hope to pursue. I plan to continue into a higher degree focused on Geographic Information Systems, and this project gives me the chance to show off my skills in advanced spatial analysis, data integration, and interpretation skills that reflect my specific interests within the field.

This project also will help me become more knowledgeable about Mars as science in the field moves closer and closer to human exploration of the red planet. New data comes in every single day, so it is a treasure trove of discoveries and conclusions to be made. My project is relevant to the field and will make me stand out, as I am confident that coming out of my Bachelor's degree with such a great understanding of the research I plan to do post-grad will give me a big leg up.

My project also gives me the opportunity to explore more of the biology side of astrobiology. Over this coming semester, I will be working with Dr. Freeland on his unpublished manuscript on a xeno amino acid alphabet that can be organized into a protein with secondary structure. I want to implement this concept into my project, as I find it the coolest thing in the entire world and it is so game-changing (at least in my opinion) that they came to the conclusion that they did, as it suggests life could be built from completely different building blocks.

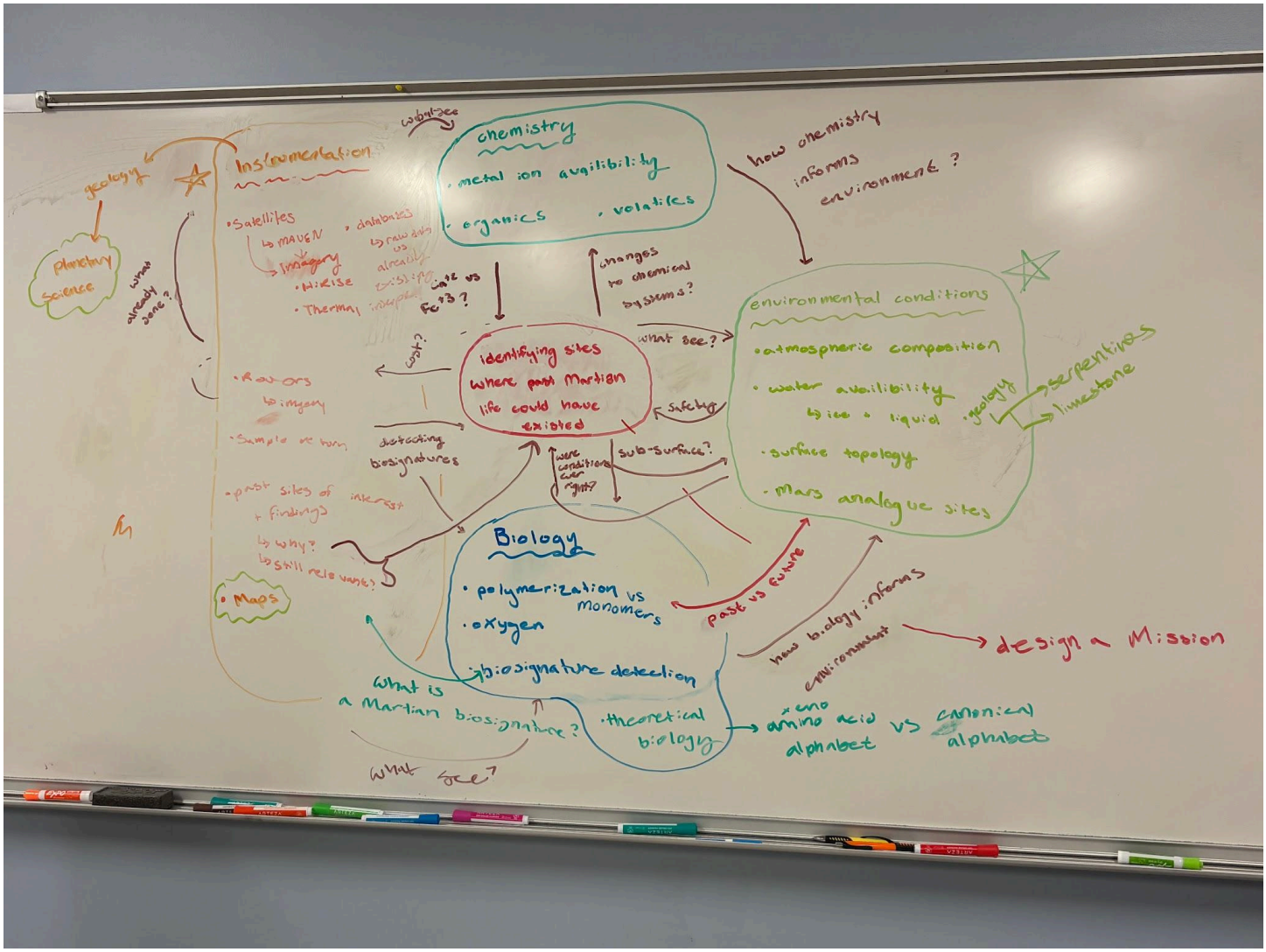
Appendix 3: Integrative Strategies

In approximately 300 words, describe, and cite any specific references to interdisciplinary [5]or integrative work in your literature. Pay attention to keywords such as *interdisciplinary, cross disciplinary, multidisciplinary, integrative, hybrid, or translational* and note these sections of your literature. How are these sections bringing perspectives (disciplines) together?

Appendix 4: Concept Map

Provide a photo of your concept map based upon the in-class concept mapping session:

[5] See Miller&Boix Mansilla (2004) for a definition of interdisciplinary work as well as descriptions of five "cognitive bridging strategies."


geology
Planetary Science
what already done?
Instrumentation
what see
•Satellites
to MAVEN
Imagery
•MRISE
•Thermal
• databases
to raw data vs already existing
chemistry
• metal ion availibility
• organics
• volatiles
how chemistry informs environment?
changes to chemical systems?
Identifying sites where past Martian life could have existed
what see?
cost?
safety
environmental conditions
•atmospheric composition
• water availibility
to ice • liquid
•geology
serpentines
limestone
• surface topology
• mars analogue sites
•Rovers
to imagery
•Sample return
detecting biosignatures
were conditions ever right?
sub-surface?
•past sites of interest + findings
to why?
to still relevant?
Biology
• polymerization vs monomers
•oxygen
•biosignature detection
past vs future
how biology informs environment
design ~ Mission
• Maps
What is a Martian biosignature?
•theoretical biology
*emo amino acid alphabet
vs
canonical alphabet
What see?

Appendix 5: Capstone Rubric

Rubric Example 1

Rubric Example 2